

P425/1
Pure Mathematics
March 2007
3 hours

Uganda Advanced Level Examinations

The paper consists of 8 compulsory questions in section A and 5 compulsory questions in section B.

All questions in the respective sections carry equal marks.

Answer all questions in both sections.

SECTION A:

1. A chord to the curve $y = x^2 - 2x + 5$ joins the points whose x – coordinates are 1 and 3. Show that the equation of the tangent parallel to the chord is $2x - y + 1 = 0$.

2. Find the perpendicular distance from the point $(1, -1, 0)$ to the plane cutting off equal intercepts on the co-ordinate axes

3. Solve $6\left(\frac{2x}{x-1}\right)^{\frac{1}{2}} + 5\left(\frac{x-1}{2x}\right)^{\frac{1}{2}} = 13$

4. Find the maximum value of $y = \frac{1}{2}x - \sin x$ in the range $0 \leq x \leq 4\pi$.

5. Prove that $\tan^{-1}(1/2) + \tan^{-1}(1/5) + \tan^{-1}(1/8) = \pi/4$

6. Prove that $7^n - 3^n$ is divisible by 4 for all natural numbers.

7. Find the sum to infinity of the G.P $\sin 2x - \sin 2x \cos 2x + \sin 2x \cos^2 2x + \dots$

8. The number of applicants for a job is 15.

- (i) Determine the number of ways in which six applicants can be selected for interviews.
- (ii) Of the 6 applicants selected 3 have background in education, 2 in business and 1 in telecommunication. Calculate the order in which they can be interviewed when applicants having the same background are interviewed successively.

SECTION B: 60 MARKS

9. (a) The curve $y = x^3 + ax + b$ cuts the x – axis at A(1,0) and B(2,0) and C. Find the co-ordinates of C.
- (b) The points A(4,0), B(0,3) and P(x, y) are such that PA is always perpendicular to PB. Determine the locus of P and give a geometrical interpretation of your answer.
10. (a) Determine the equation of the plane containing the line $\frac{x-1}{3} = \frac{y-1}{2} = \frac{z+2}{2}$ but does not contain the point (4,0,3).
- (b) A line L passing through the point (2,4,3) is perpendicular to the plane in (a) above; Find the:
- (i) the equation of L,
- (ii) point of intersection of L and the plane in (a) above.
11. (a) Show that when the equations $x^2 + \alpha x + \beta = 0$ and $x^2 + \lambda x + \gamma = 0$ have a common root, then, $(\beta - \gamma)^2 = (\alpha - \lambda)(\beta\lambda - \alpha\gamma)$
- (b) Given that $p(x) = 8x^3 - 12x^2 - 18 + k = 0$ has a repeated root, find the possible values of k.
12. (a) Given that $y = a\cos^2\theta + b\sin^2\theta$, show that $\frac{d^2y}{dx^2} + 4y = 2(a + b)$
- (b) A right circular cone of vertical angle 2α is inscribed in a sphere of a fixed radius a with its vertex and rim of its base on the sphere. Show that the volume of the cone is $\frac{8}{3}\pi^3\sin^2\alpha \cos^4 \alpha$ and hence find its maximum value, stating the value(s) of α for which it occurs.
13. (a) From a point A, due southeast of a tower, the elevation of the top of the tower is observed to be 32° . From A the observer walks in a straight line 100° west of North for 320 metres and finds that he is then southwest of the tower. Assuming that the foot of the tower and the line of the observers' walk are in the same horizontal plane, find the height of the tower.
- (b) Solve $2\sin 2\theta - \sin \theta - 12\cos \theta + 3 = 0$ for $0^\circ \leq \theta \leq 360^\circ$
-
-